SPARKY'S SCIENCE FACTS

# How Do Bridges Stay Up?

Engineering · Ages 7+

> **Short answer:** Bridges stay up because engineers choose strong shapes, like triangles and arches, that send the weight safely down into the ground.

## How it works

- A beam bridge is a strong beam resting on supports at each end, like a plank over a stream.
- An arch bridge curves upward. The arch pushes the weight sideways and down into the ground at its ends.
- A suspension bridge hangs from strong cables held up by tall towers.
- Triangles are used in many bridges because a triangle keeps its shape when it is pushed. A square can be squashed into a diamond.

**Key words:** Load: the weight a bridge carries; Truss: a frame made of triangles

## Common mix-ups

- Thicker is not always stronger: the shape matters. A folded sheet of paper holds much more than a flat one.
- Bridges are not perfectly still. They bend a little, safely, under heavy loads and wind.

**Did you know?** The Bloukrans Bridge on the Garden Route is a huge concrete arch bridge. People bungee jump 216 metres from it!

**Try it:** Lay a sheet of paper between two books and put coins on it. Then fold the paper like a fan and try again. Which holds more?

**Ask Sparky:** Why are there triangles on cranes? How do bridges float on water?